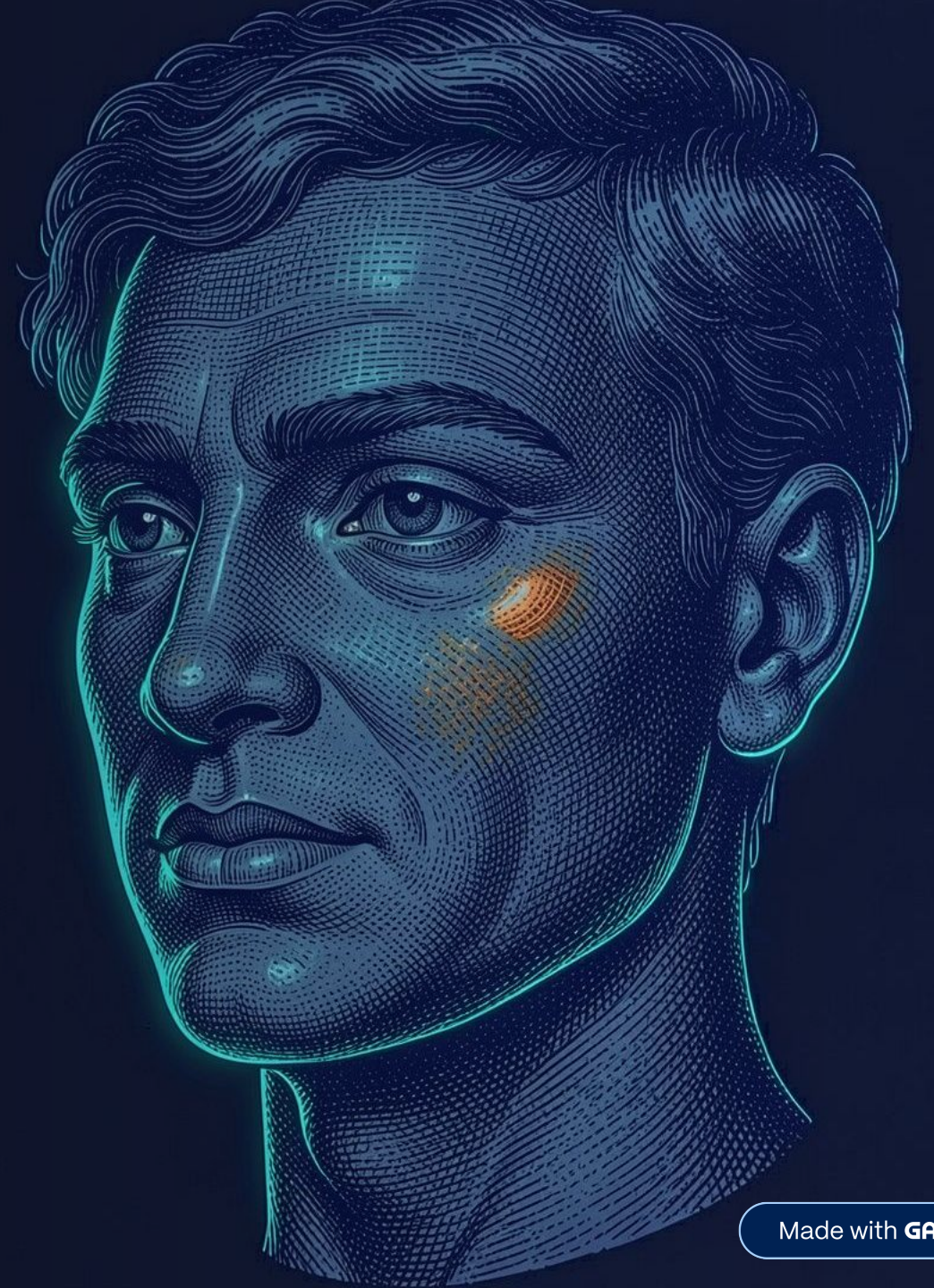


DEEP LEARNING • COMPUTER VISION

Face Expression Recognition Using CNN

A complete deep learning pipeline for
classifying seven human facial
expressions from images.

SHAFAT ULLAH • GOOGLE COLAB • TENSORFLOW /
KERAS



Made with GAMMA

THE TASK

Turning facial pixels into expression labels

- Build a CNN-based computer vision system that recognizes Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise.
- Cover the full lifecycle: dataset preparation, training, evaluation, prediction, and model reuse.



END-TO-END PIPELINE

The project follows an end-to-end learning pipeline

Each stage turns raw images into a reusable classifier.

01

Explore

Download and explore the dataset.

02

Prepare

Preprocess images and augment training examples.

03

Train

Construct, compile, and train the CNN.

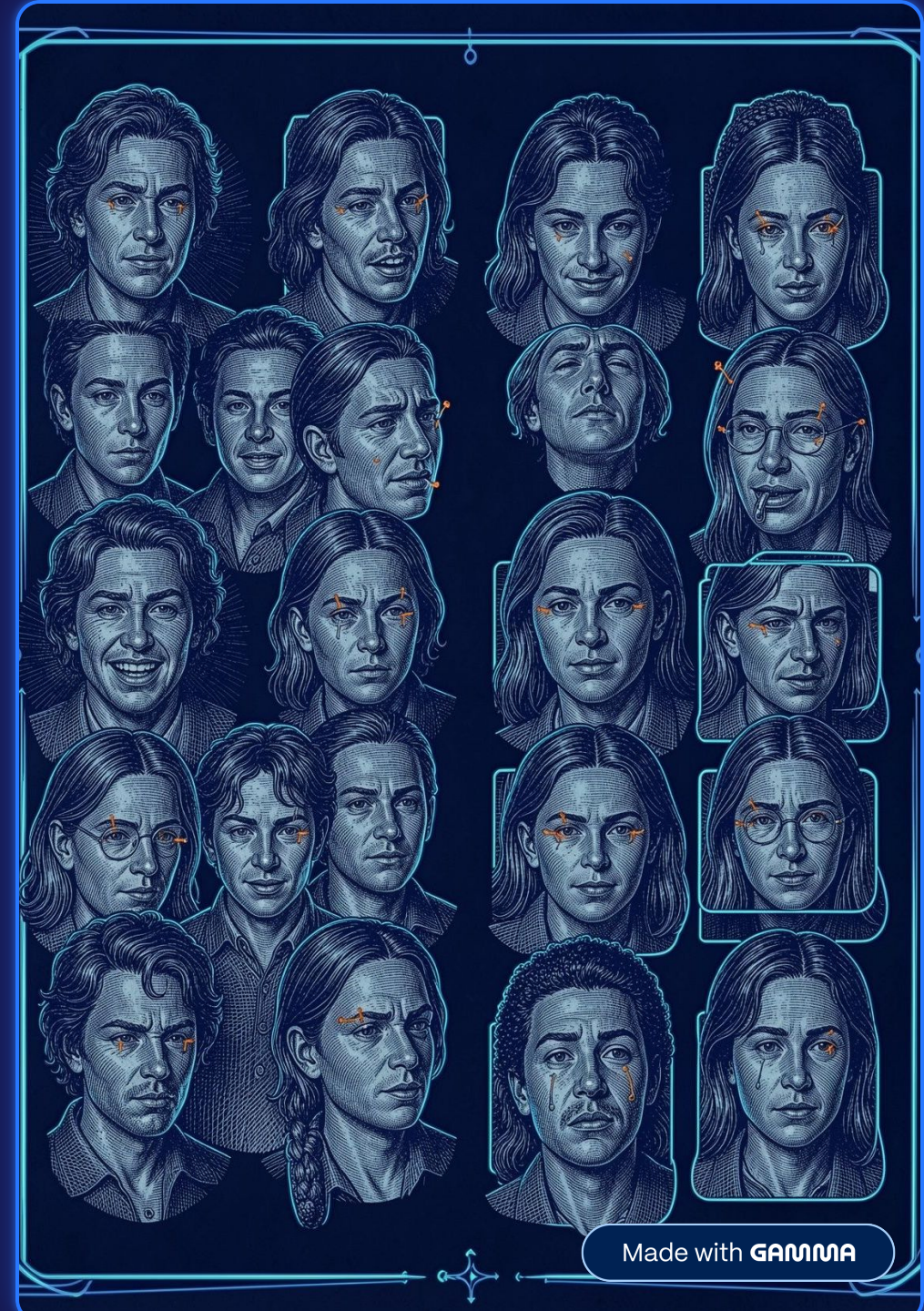
04

Reuse

Evaluate errors, predict new images, and save the best model.

The dataset is organized for supervised seven-class learning

- Face Expression Recognition Dataset accessed through KaggleHub in Google Colab.
- Separate train and validation folders, each organized by the seven expression categories.
- Dataset source: Jonathan Oheix, Face Expression Recognition Dataset on Kaggle.



Preprocessing makes every image comparable



Standardize

Resize images to 48×48 pixels; convert to grayscale for a $48 \times 48 \times 1$ input.



Augment

Augment training images with rotation, width and height shifts, zoom, and horizontal flipping.



Normalize

Normalize pixel values from approximately 0–255 to 0–1; normalize validation images without augmentation for a consistent evaluation baseline.

MODEL ARCHITECTURE

The CNN learns from simple features to expressive patterns

Four convolutional stages move from low-level edges and curves to richer facial patterns, followed by a dense classifier and a seven-class Softmax output.

Input 48 × 48 × 1	Conv2D 32 filters	Conv2D 64 filters	Conv2D 128 filters
Conv2D 256 filters	Flatten Feature vector	Dense 256 units	Dense 7 Softmax output



Each CNN component has a specific role



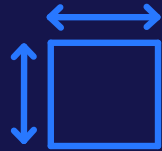
Convolution

Extracts facial features: edges, eyes, eyebrows, and mouth patterns.



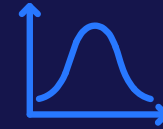
Batch Normalization

Stabilizes training.



Max Pooling

Reduces spatial dimensions, retaining important information.



Activation and regularization

Dropout helps reduce overfitting; ReLU activates hidden layers and Softmax produces class probabilities.

TRAINING CONFIGURATION

Training is configured for stable optimization

64

batch size

10

maximum epochs

0.0005

learning rate

Adam optimizer · Categorical crossentropy supports the seven-class classification task.

EarlyStopping

Can end training when validation accuracy stops improving.

ReduceLROnPlateau

Adjusts learning rate after validation loss plateaus.

ModelCheckpoint

Saves the best validation-accuracy model.

EVALUATION

Evaluation looks beyond a single accuracy number

Validation loss and accuracy

Summarize overall performance.

Confusion matrix

Reveal which expressions are commonly confused.

Precision, recall, F1-score, and support

Show class-level behavior.

Actual-versus-predicted images

Provide a visual check of model behavior.

No numerical results are assumed.

INFERENCE AND REUSE

The trained model can recognize new images

1. Upload a new face image in Google Colab.
2. Resize, grayscale, and normalize it using the same preprocessing pipeline.
3. Pass it through the CNN and select the highest Softmax probability.
4. Interpret the probability as model confidence, not a guarantee of real-world correctness.

face_expression_cnn.keras — The trained model is saved for reuse without retraining.

